Physics 5A Discussion (Week 2)

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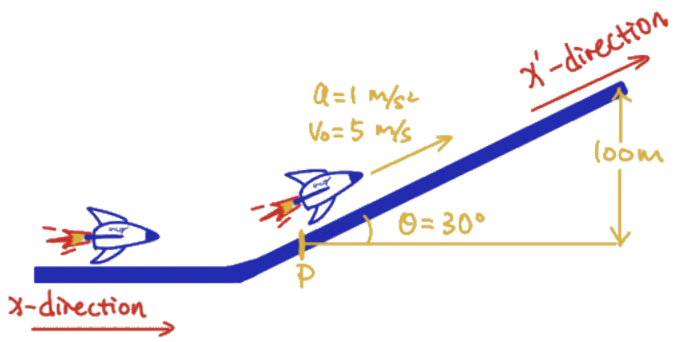


Problem 1: After accelerating to the right for a few seconds, the rocket moves onto a ramp at an angle of $\theta = 30°$. When it reaches point P as shown, it has an instantaneous velocity $v_0 = 5$ m/s in the x'-direction, and its acceleration in the x'-direction is kept constant at 1 m/s^2. How long does it take for the rocket to reach a height of $h = 100$ m? This is quite a complicated problem, but we can break it down into a few steps:

Step 1: Looking at the triangle shown, how far does the rocket need to travel in the x'-direction (distance on the ramp) to reach $h = 100$ m?

Step 2: Draw the acceleration vs. time graph, with the acceleration in the x'-direction on the vertical axis, and time on the horizontal axis. *(Hint: As mentioned in the problem, acceleration is kept constant over time.)*

Step 3: Using the graph you have in step 2, draw the velocity vs. time graph for the first 20 seconds. (*Hint: Change in velocity is the "area" under the acceleration curve.*)

Step 4: Using the graph you have in step 3, what is the shape of curve in your position vs. time graph? (Hint: The slope of your position curve at each time instant should be the value of your velocity curve at the same time instant. Alternatively, the change in the area under the velocity curve corresponds to the amount of increase/decrease in the position curve.)

Step 5: Again using your velocity vs. time graph, how far has the rocket traveled in the x'-direction in the first 20 seconds? (*Hint: Use the "area" under the velocity curve to find displacement.*)

Step 6: Recall the distance calculated in step 1, how long does the rocket take to travel such a distance? (*Hint: Try an arbitrary final time t_f and repeat from step 3. You will need the quadratic formula when computing your final answer.*)

Step 7: Now you have seen that *position curve looks like a parabola if we have a constant acceleration over time*, this result leads to the kinematic equation that you will see very soon during lecture:

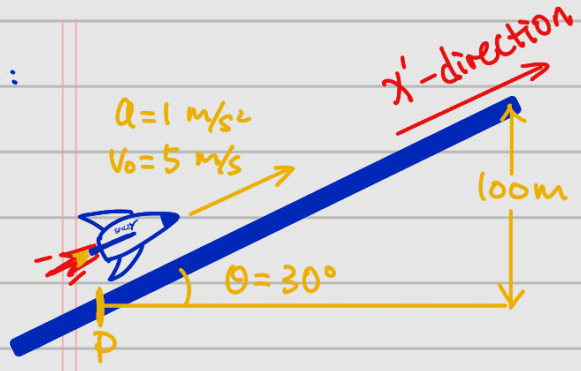
$$x = \frac{1}{2}at^2 + v_0 t + x_0 \,,$$

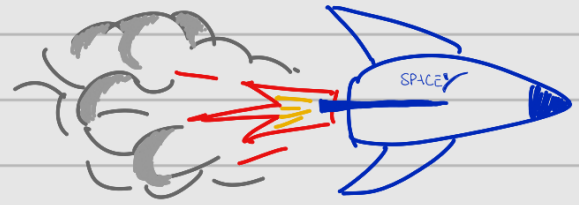
where x_0 is your initial position at time $t = 0$ second, and x is your final position at a generic time t. Using this formula, find the time t for the rocket to travel from $x_0 = 0\,\mathrm{m}$ to $x =$(your result in step 1) in the x'-direction. (*Hint: You will need the quadratic formula. The result you get here should agree with the one you found in step 6.*)

Problem 2: For the rocket in problem 1, how far has the rocket traveled in the horizontal direction (x-direction) when it reaches $h = 100$ m?

Review the week-1 worksheet (rocket moving left and right) if you have time :)
Especially if you want some practice on "average speed" and "average velocity".

Setup:





Step 1. the rocket is travelling on the incline (hypotenuse)
We want it to go up by 100 m (opposite side)
↳ trigonometry !!

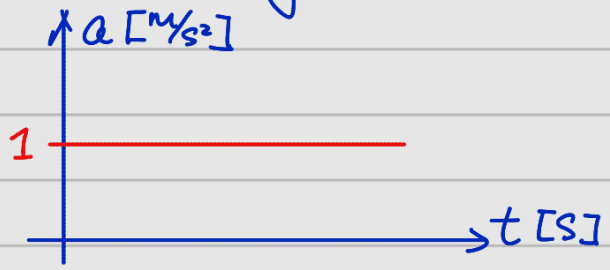
$$\sin(30°) = \frac{100\ m}{\Delta x'}$$

$\sin(30°) = \frac{1}{2}$

$\Delta x'$ is displacement on the incline

so $\boxed{\Delta x' = 200\ m}$

Step 2 acceleration is kept constant $a = 1\ m/s^2$ over time
↳ flat straight line



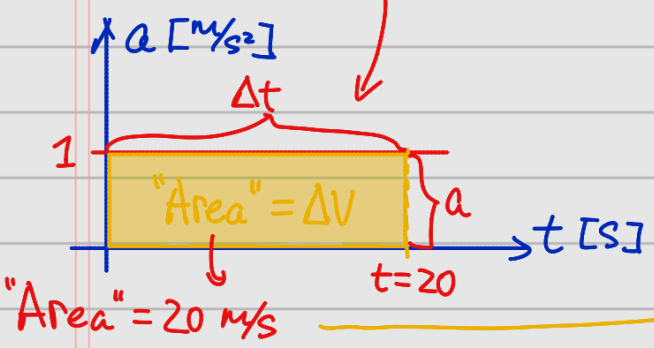
Step 3 remember $a = \lim_{\Delta t \to 0} \frac{\Delta V}{\Delta t}$ if a is constant, ignore "$\lim_{\Delta t \to 0}$"
↳ $\Delta V = a \cdot \Delta t$ for constant acceleration.

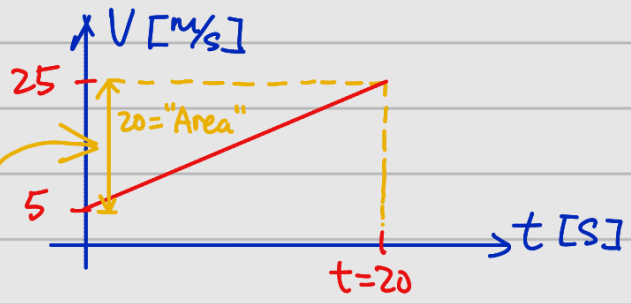
Change in velocity $\Delta V = V_{final} - V_{initial}$

You pick a final time and this is instantaneous velocity at that time. In this case we pick $t_{final} = 20\ s$

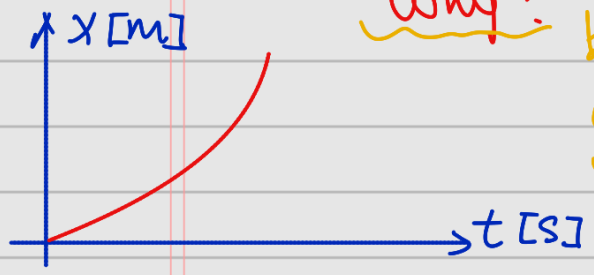
this is $V_0 = 5\ m/s$ at $t = 0\ s$ (given in the problem)

So $\Delta V = a \cdot \Delta t = (1\ m/s) \cdot (20\ s - 0\ s) = 20\ m/s$



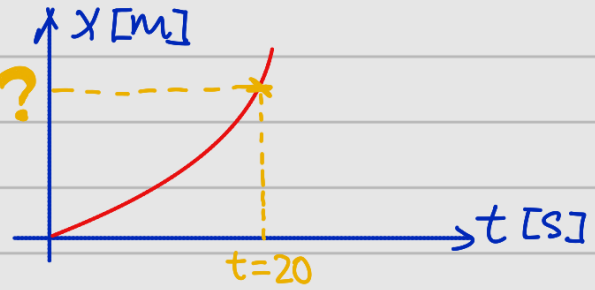


Step 4 As you have seen many times (week-1 worksheet) when velocity is a tilted straight line, position curve should be a parabola, more specifically, v is growing in this problem, so the parabola is concave-up. Since v is always larger than 0 in this problem, we only have the right-half of the parabola.



Why? because rocket is moving faster and faster to the right
↪ x grows faster and faster over time, never decreases

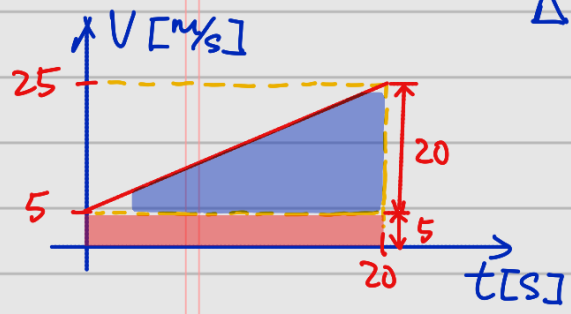
Step 5 We want to find this number →



to do this, recall what we have

$\Delta x'_{0\to 20}$ = "Area" under v from 0 s to 20 s

displacement from t=0 to t=20 s
(or change in position)



$\Delta x'_{0\to 20}$ = "Area" under v from 0 s to 20 s

= Area of ◢ + Area of ▬

$= (20\,s)\cdot(25\,m/s - 5\,m/s)/2 + (20\,s)\cdot(5\,m/s)$

$= 200\,m + 100\,m = \boxed{300\,m}$

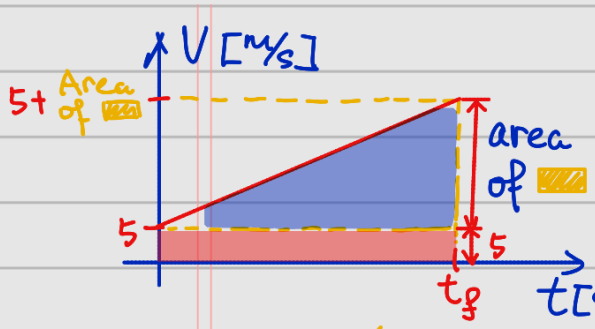
So the rocket has traveled 300 m in 20 seconds, which is more than the 200 m (Step 1) that we want for it to go up 100 m vertically.
↪ it takes less than 20 s for it to travel 200 m on the ramp

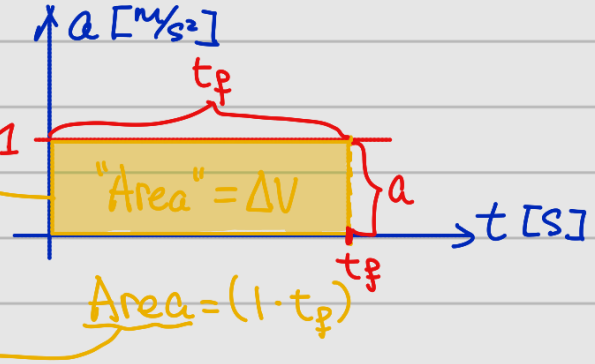
Step 6 For it to travel 200 m on the ramp, set $\Delta x'_{0\to ?} = 200\,m$

$\Delta x'_{0\to ?}$ = "Area" under v from 0 s to ? s = 200 m

= Area of ◢ + Area of ▬

(see pictures on next page)





Area $= (1 \cdot t_f) \cdot t_f/2$

Area $= 5 \cdot t_f$

$\Delta x'_{0 \to t_f}$ = "Area" under v from 0 s to t_f s = 200 m
= Area of ◢ + Area of ▬

$$200 = (1 \cdot t_f) \cdot t_f/2 + 5 \cdot t_f$$

quadratic equation for t_f

Solve for t_f using quadratic formula

$$t_f = \frac{5^2 \pm \sqrt{5^2 - 4(\frac{1}{2}) \cdot (-200)}}{2 \cdot (1/2)} = \boxed{15.6\,s}$$

keep only positive t

time it takes for it to go up 100 m vertically.

Step 7

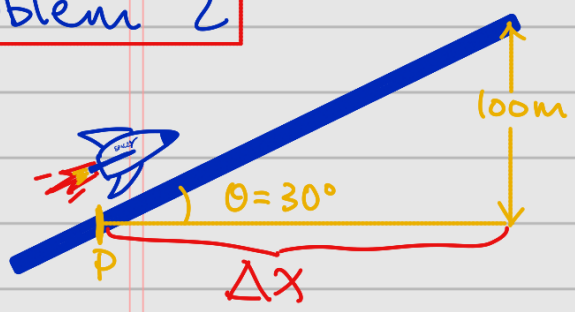
$200 = (1 \cdot t) \cdot t/2 + 5 \cdot t$ ← equ we get in step 6

$x - x_0 = a\, t^2/2 + V_0\, t$ ← formula in step 7

we have just solved for this quadratic equ. t = 15.6 s.

* This is where the formula comes from.
* In the future, if you see similar problem again with constant acceleration, apply the formula directly, no need to calculate the "areas" again.

Problem 2



trigonometry !!

$$\tan(30°) = \frac{100\,m}{\Delta x}$$

$$\Rightarrow \Delta x = \frac{100\,m}{\tan(30°)} = 100\sqrt{3}\,m$$